From Molecular Dynamics in Virtual Reality Environments to Fundamental Questions about Human-Computer Interaction

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ABSTRACT

By nature, the molecular systems that are of interest to structural biologists exhibit high spatial and temporal complexity, in part due to the sheer density of objects and their constant motion. Virtual reality is a very useful tool to manage this complexity and enable interactive simulations that would otherwise be impractical. However, current solutions provide no adequate way to select very small objects whose movements are fast and unpredictable, and theoretical tools such as Fitts' law are of no help with unpredictable motion. To guide the design of appropriate selection techniques, notably with haptic assistance, we studied how subjects performed when attempting to select such objects in a simpler, 2D environment. We propose a new theoretical framework to model selection performance as a function of the predictability of the moving targets.

1 Molecular Dynamics and Selection

In a molecular dynamics [7] (MD) simulation, a complex molecular system composed of up to hundreds of thousands of atoms is modeled and simulated to better understand its dynamics and the corresponding biologically-relevant phenomena. Often, in such simulations, every atom is visually represented, usually by a sphere. This leads to extremely cluttered environments and confused shapes further obfuscated by the constant motion of their constituent parts.

By allowing structural biologists to immerse themselves into these complex molecular systems, virtual reality facilitates their understanding. It is perhaps even more useful when MD simulations are designed to be interactive [8], i.e. to allow their users to select objects (e.g. atoms, amino acids or secondary structures) and to apply forces that take effect in the simulation. This is usually achieved with a haptic device, so that biologists can first "feel" the molecular system when exploring it, but also test its flexibility and measure its forces when they manipulate it.

However, MD environments are visually very rich and dense, so that most objects are at least partially occluded. More inconvenient still is the fact that all of these small objects move very fast, and that their movements cannot be predicted. The combined effect of these circumstances is that selecting any object tends to be very difficult, if not impossible unless undesirable measures are taken. The latter may include drastically slowing the simulation down, or applying visual smoothing filters that would result in a loss of information.

2 Fitts' Law and Theoretical Shortcomings

Selection has been a major area of focus in Human-Computer Interaction for a long time, and of experimental psychology for longer still. One of the most important findings in this field is known as Fitts' law [2], often formulated by equation 1. In the latter, T is the average time of selection, a and b are constants, D is the distance from the starting point to the center of the target and W is the width of the target.

$$T = a + b \times log_2\left(1 + \frac{D}{W}\right) \quad (1)$$

Fitts' law, which was originally established for one-dimensional selection, has been extended to two [5] and three [6] dimensions, to account for obstacles along the path to the target [1], or even for moving targets [3], [4]. However, to our knowledge, the nature of the targets' movements, in particular as it relates to predictability, has never been studied.

3 Formalization of Irregular Motion

To address this issue, we designed a formal way to generate irregular motion of quantifiable irregularity. We characterize the nature of motion according to three parameters:

- S is the speed at which targets move, in cm/s;
- F is the frequency at which every target's direction vector is rotated, in Hz;
- A is the maximum angle by which the direction vector of a target can be rotated. More precisely, after every period $T = \frac{1}{F}$, the direction vector of the target is rotated by an angle α pseudo-randomly sampled from $[-A;+A]$. A is in degrees of arc.

We designed an experiment in which subjects were asked to click on a moving target with a mouse, using a desktop monitor. The task was performed in a square window, 23.87cm and 1000 pixels on each side. In this window 50 disks were displayed; 49 of these were gray, and one was red; the latter was the target the subjects were to select, the former were distractors.

All disks moved according to the S, F and A parameters introduced above. We tested every combination of the following values: S: 0.73, 1.46, 2.19 cm/s; F: 1, 2, 4, 8, 13, 20, 30 Hz; A: 0, 30, 60, 90, 120, 180 degrees of arc.

We also added a condition where S = 0, i.e. where targets were static. For every condition, a block of 4 successive trials was tested.

Selection always involves a trade-off between speed and accuracy, and different people can spontaneously adopt different approaches to this trade-off. In order to minimize the effect of such variation in selection strategy, we instructed our subjects to try to select their target using 5 clicks or (preferably) fewer. If they failed to do so, we instructed them to keep trying to select the target anyway, but without minding errors. Thus, failures are not absolute, but indicate critical conditions for which selection is very difficult.

4 Results and Conclusion

The heat maps in figures 1 and 2 present our results for the highest speed we tested, because they are the most telling. In each of these, angle widths are represented on the horizontal axis. More precisely, the value 90 stands for conditions where direction vectors were rotated by an angle between -90 and +90 degrees. Frequencies are

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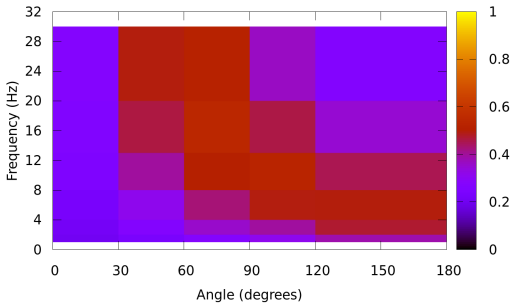


Figure 1: Heat map of average normalized selection times.

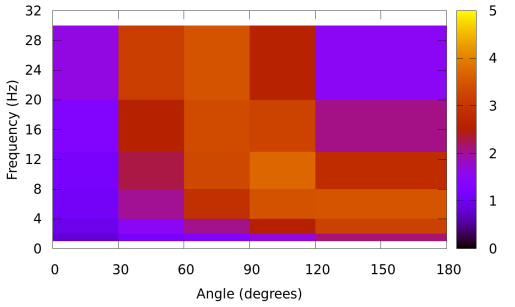


Figure 2: Heat map of average error rates, out of a maximum of 5.

represented on the vertical axis. Thus each intersection is the average of our results for the corresponding condition.

Figure 1 displays the average of normalized selection times. We observe that a region of the heat map approximately centered on a diagonal (from top-left to bottom-right) exhibits the highest selection times. Figure 2 displays the average error rates for each condition, i.e. the number of clicks that missed the target; note that missed clicks after the fifth one were not taken under account. The results are very strongly correlated with selection times.

The heat maps described above indicate that selection is especially difficult for combinations of wide angles with low frequencies; narrow angles with high frequencies, or medium angles with moderate frequencies. This suggests that the product AF of angle width and frequency might reliably predict selection performance for targets of irregular motion. To better illustrate this, we plotted this product against selection times in figure 3. Based on these preliminary results, it would appear that an index of difficulty (or predictability) for irregular motion could be devised based on the difference between AF and a reference value, which in this case is around 1,100. Future work should examine how this constant relates to parameters that were left untouched in our study, such as target width or distance. Based on a questionnaire we submitted to our subjects after the experiment, it seems that very high values of AF are relatively easy because the corresponding targets are perceived as "vibrating" around a somewhat stable average position.

Our hope is that the theoretical framework sketched here in a 2D context will provide valuable insight to guide the design of effective selection techniques specifically tailored to moving targets in virtual environments, especially targets whose movements are difficult to predict, as in MD simulations. However, we still need

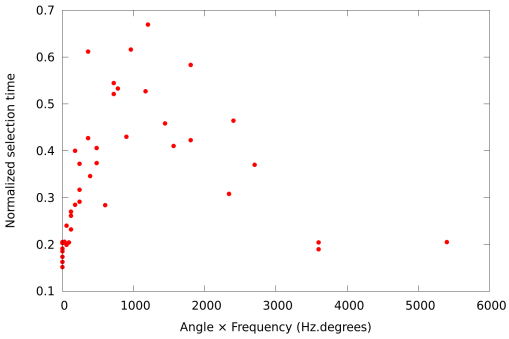


Figure 3: Average selection times vs. predictability.

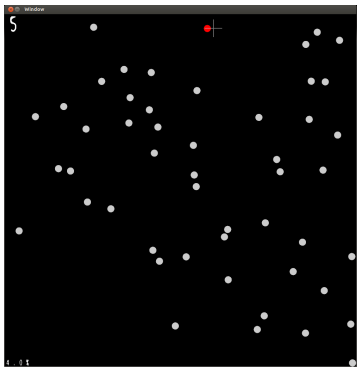


Figure 4: Application used for the experiment.

to investigate whether our results hold well when targets move in 3D spaces where occlusion (minimal in our study) could be high, when stereoscopic rendering is added, or when different input devices are used. Nonetheless, we expect that a combination of target-prediction heuristics and haptic feedback would yield good results.

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